



KOLEJ MATRIKULASI MELAKA
UNIT MATEMATIK
SEMESTER I, SESI 2022/2023
MATHEMATICS 1 SM015
KARISMATIK A



ANSWER ALL QUESTIONS

PART A

1. Find the following limits :

a) $\lim_{x \rightarrow \infty} \frac{(1+x)(2+x)}{x^2+1}$ (4 marks)

b) $\lim_{x \rightarrow 1} \frac{x-1}{\sqrt{5x-4}-\sqrt{x}}$ (4 marks)

2. Find $\frac{dy}{dx}$ for each of following:

(a) $y = \frac{x^3}{x^3-1}$ (b) $y = e^{3x} \ln(2x)$ (8 marks)

3. Given that $f(x) = x^3 - 2x^2 + x + 1$

- a) Use the second derivative test to determine the relative extremum of the function.
b) Find the point of inflection. (9 marks)

PART B

1. Express $\frac{z\bar{z}-5i}{2+i}$ in the form $a+bi$ where $z = -1+3i$ and $\bar{z}$ is a conjugate of z . Hence

find $\left| \frac{z\bar{z}-5i}{2+i} \right|$. (5 marks)

- 2 (a) Solve the equation $10 \log_x 3 - \log_3 x = 3 \log_2 8$. (5 marks)

- (b) By writing $|x| \geq a$ in the equivalent form $x \geq a$ or $x \leq -a$, determine the

solution set for $\left| \frac{x-3}{2x-1} \right| \geq \frac{1}{2}$. (6 marks)

3. Given $f(x) = \sqrt{2x+3}$, and $g(x) = \frac{1}{2}(x^2-3); x \geq 0$.

- (a) Find domain and range of $f(x)$ and $g(x)$.

- (b) Show $g(x)$ is the inverse of $f(x)$ or vice versa. (8 markah)





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4. The functions f and g are defined as $f(x) = \ln(x+2)$, $x > -2$ and $g(x) = \frac{x}{2}$
- (a) Find $f^{-1}(x)$ and $g^{-1}(x)$
- (b) Evaluate $(g^{-1} \circ f^{-1})(2)$.
- (c) If $(f \circ g)^{-1}(k) = \frac{1}{2}$, find the value of k . (12 marks)
5. Sand is added at the rate of $12\pi \text{ cm}^3 \text{ s}^{-1}$ into an inverted cone with a semi-vertical angle 30° . Find
- a) the rate of increase in the depth of the sand when the sand level is 6cm.
- b) the time that has elapsed when the depth of the sand is 9 cm. (9 marks)
6. a) Determine the horizontal asymptotes for the following functions, $f(x) = \frac{\sqrt{9x^2 - 1}}{1 - x}$
- (5 marks)
- b) Given that $g(x) = \begin{cases} \frac{x^2 + 5|x|}{2x}, & x \neq 0 \\ \frac{5}{2}, & x = 0 \end{cases}$
- i) Determine whether $g(x)$ is continuous at $x = 0$
- ii) Determine whether $g(x)$ is continuous from the right or left at $x = 0$
- (9 marks)
7. (a) The equation of a curve is given by $y^3 + 2y = \sin x + 3$. Find the gradient of the curve at the point (0,1). (4 marks)
- (b) Given the parametric equation $x = t - \frac{2}{t}$ and $y = t + \frac{2}{t}$ where $t \neq 0$.
- (i) show that $\frac{dy}{dx} = 1 - \frac{4}{t^2 + 2}$ (ii) Find $\frac{d^2y}{dx^2}$ when $t = 1$
- (12 marks)

END OF QUESTIONS

